

The objectives of this lab are to create a parallel plate capacitor, measure its capacitance for variations of area, plate separation and dielectric material, and compare our results to theoretical expectations.

A *parallel plate capacitor* consists of two metal plates separated by an insulating material. The plates have area A , they are separated by distance d , and the insulating material has a *dielectric constant* abbreviated as κ . The *capacitance* of the capacitor is given by the expression:

$$C = \frac{\epsilon_0 \kappa A}{d}$$

The dielectric constant, κ , is a unitless number, typically less than 5 (for common materials). The units of ϵ_0 are:

$$C^2 / N \cdot m^2$$

If area A and separation d are measured in m^2 and m , respectively, then the units of capacitance will be:

$$C^2 / N \cdot m \quad \text{which happens to be equivalent to a } \textit{Coulomb} / \textit{Volt}.$$

We call this unit a *Farad*, abbreviated F , in recognition of the contributions of Michael Faraday.

This also means that we can express the units of ϵ_0 using *Farad*:

$$C^2 / N \cdot m^2 \text{ is } C^2 / N \cdot m \text{ divided by } \textit{meter} \text{ or } \textit{Farad} / \textit{meter}$$

So our value of ϵ_0 can be written as:

$$8.85 \times 10^{-12} C^2 / N \cdot m^2 \text{ will become } 8.85 \text{ } \mu\text{bi-Farad} / \textit{meter}$$

Preliminary Measurements

- Measure the dimensions of one metal plate, with L the longer dimension and w the shorter dimension. Record these in **meters**.
- Calculate the thickness of one piece of paper in units of **$\times 10^{-3}$ meters**.

Part 1: vary d while holding A constant

- Place one sheet of white paper between the metal plates. Align the plates so their areas perfectly overlap. Clamp the plates together, using blocks where needed. Measure and record the capacitance, in **pF**.
- Repeat the first step for two through ten sheets of paper.

- Create a data table with columns for:

of sheets d (mm) d^{-1} (m^{-1}) C (pF)

- Create a graph of C vs d^{-1} and find the equation of the best fit line.
- Calculate the experimental value of the dielectric constant, κ , for white paper.

Note that in our original equation, we can replace A , the area of the plates, with the product of wL . The equation then becomes:

$$C = (\epsilon_0 \kappa w L) d^{-1}$$

So the slope of our graph should equal:

$$\text{slope} = (\epsilon_0 \kappa w L)$$

We can calculate the dielectric constant by rearranging this expression:

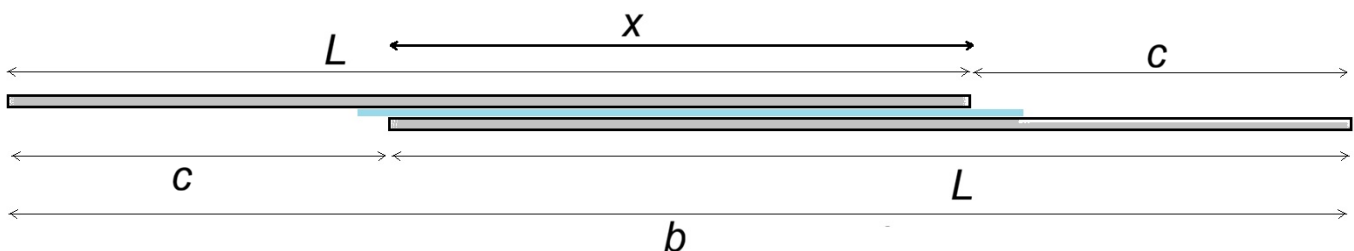
$$\kappa = \frac{\text{slope}}{\epsilon_0 w L} = \frac{(\dots \text{pF-m})}{(8.85 \text{ pF/m})(\dots \text{m})(\dots \text{m})} = \dots$$

Notice that if we use compatible units for the values in the calculation, κ will be a unitless number.

Part 2: vary A while holding d constant

Place one sheet of white paper between the metal plates and align the plates so that their **widths overlap**. In Part 2, we will slide the plates – varying their common length – in order to vary the common area of the plates.

If we define the total length of the plates – from the end of one plate to the end of the other – as “ b ”, then we can define “ x ” as the length they have in common, as shown here:



In this diagram, the gray rectangles represent the side view of the metal plates and the blue represents the dielectric between them. We can see that:

$$b = L + c \quad \text{and} \quad c + x = L$$

By adding these two equations together and simplifying the result, we get:

$$x = 2L - b$$

Using this expression, we can measure b , the total length of the two plates, and calculate the corresponding value of x , which will relate to the area of the capacitor.

- Arrange the plates so that 2 cm of one is showing (i.e. so that “ c ” in the diagram above is 2 cm). Clamp the plates together, using blocks where needed. Carefully measure and record the value of b , in meters, and the capacitance, in μF .
- Repeat the first step for values of “ c ” of 4 cm, 6 cm, ... , 18 cm, 20 cm.
- Create a data table with columns for:

b (m) x (m) C (μF)

- Create a graph of C vs x and find the equation of the best fit line.
- Calculate the experimental value of the dielectric constant, κ , for white paper.

Note that in our original equation, we can replace A , the area of the plates, with the product of w and x . The equation then becomes:

$$C = \left(\frac{\epsilon_0 \kappa w}{d} \right) x$$

So the slope of our graph should equal:

$$slope = \left(\frac{\epsilon_0 \kappa w}{d} \right)$$

We can calculate the dielectric constant by rearranging this expression:

$$\kappa = \frac{d (slope)}{\epsilon_0 w} = \frac{(\dots m)(\dots \mu F/m)}{(8.85 \mu F/m)(\dots m)} = \dots$$

Notice that if we use compatible units for the values in the calculation, κ will be a unitless number.

Repeat Parts 1 and 2 for the other two dielectric materials, mylar and waxed paper.